## ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are true, but R does not explain why the image is inverted. |
| 2 | **(c)** -25 cm | f = -(far point) = -25 cm. |
| 3 | **(b)** smaller wavelengths are scattered more by small particles | Rayleigh type scattering. |
| 4 | **(c)** A is true but R is false. | A is true; R is false: the image forms in front of the retina. |
| 5 | **(b)** -1 D and +1.5 D | Distance: f = -100 cm. Reading: 1/f = -1/40 + 1/25 = 3/200, P = +1.5 D. |
| 6 | **(a)** less red than in the plains at the same time | Thinner atmosphere, less scattering. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | Higher n means more bending. |
| 8 | **(b)** the ciliary muscles changing the shape of the lens | Accommodation by the ciliary muscles. |
| 9 | **(c)** larger than the wavelength of light | All colours scatter. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | Shorter wavelength scatters more. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1). | 1+1 |
| 12 | Colours always separate (1); Because n differs for each colour (1). | 1+1 |
| 13 | Iris limits the entry of light (1); Lens changes its curvature to focus (1). | 1+1 |
| 14 | 1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | 1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1). | 1+1+1 |
| 16 | Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1). | 1+1+1 |
| 17 | At 10 cm (1); The lens must be thicker and have a shorter focal length (1); For 100 cm the lens is thinner (1). | 1+1+1 |
| | **OR:** Hypermetropia and myopia (1); Reading: 1/f = -1/100 + 1/25, P = +3 D (1); Distance: f = -2 m, P = -0.5 D (1). | 1+1+1 |
| 18 | Prism disperses the light (1); Edges appear coloured due to unequal deviations of colours (1); Violet and red are separated (1). | 1+1+1 |
| | **OR:** Sky is bright due to scattered sunlight (1); Stars' light is overpowered (1); At night no scattering (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (a) f = -90 cm, P = -1.11 D (1); (b) Concave lens (1); (c) f = -100 cm; image must be at 90 cm (1); 1/u = 1/v - 1/f = -1/90 + 1/100 = -1/900 (1); New far point is 900 cm (9 m), so correction is incomplete (1). | 1+1+1+1+1 |
| | **OR:** u = infinity, v = -2 m (1); f = -2 m (1); P = -0.5 D (1); Image at the far point, 2 m (1); It is virtual and erect for the lens (1). | 1+1+1+1+1 |
| 20 | (a) 1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1); (b) f = 100 cm, 1/u = 1/v - 1/f = -1/50 - 1/100 (1); 1/u = -3/100 (1); u = -33.3 cm, so near point 33.3 cm (1). | 1+1+1+1+1 |
| | **OR:** (a) Red scatters least, so visible from far (1); (b) Yellow/red fog lamps penetrate better (1); (c) Earth sky is blue due to scattering (1); (c) Moon has no atmosphere, so a black sky (1); Scattering depends on wavelength and particle size (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** white light | The second prism reverses the dispersion. |
| 2 | **(c)** go undeviated | No relative refraction. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | Atmospheric refraction. |
| 4 | **(c)** is nearly monochromatic | Single wavelength. |
| 5 | **(a)** violet travels slower than red | n(violet) > n(red). |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | Averaging of variations. |
| 7 | **(a)** Pass a beam through water with a few drops of milk | Colloidal milk particles scatter light. |
| 8 | **(b)** Light of all colours gets scattered many times | Multiple scattering mixes colours. |
| 9 | **(c)** A is true but R is false. | A is true; R is false: red has the longest wavelength and is scattered least. |
| 10 | **(c)** is not deviated appreciably | Normal incidence along vertical layers. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Distant rays are parallel and the eye can accommodate (1); Near objects need greater accommodation, which is lacking (1). | 1+1 |
| 12 | Hypermetropia (1); Near objects are focused behind the retina (1). | 1+1 |
| 13 | f = -60 cm (1); P = 1/(-0.6) = -1.67 D (1). | 1+1 |
| 14 | Their light passes through more atmosphere (1); Greater variation in the refraction (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Atmospheric particles are smaller than the wavelength (1); Short wavelengths scatter more, so noon sky is blue (1); Long path at sunset scatters away blue, leaving red (1). | 1+1+1 |
| 16 | Sun emits less violet (1); Upper atmosphere absorbs violet (1); Our eyes are less sensitive to violet (1). | 1+1+1 |
| 17 | Dark/black (1); Moon has no atmosphere (1); Without scattering the Sun's light isn't spread (1). | 1+1+1 |
| | **OR:** Red bends the least and returns at a larger angle (1); Violet bends the most and returns at a smaller angle (1); Hence red appears at the top (outer) and violet at the bottom (1). | 1+1+1 |
| 18 | Tiny droplets scatter the light (1); Scattering spreads light in all directions (1); Light seems to surround the source (1). | 1+1+1 |
| | **OR:** In the dark more light is needed (1); In bright light the pupil shrinks to protect the retina (1); Controlled by the iris muscles (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | -2 D means f = -50 cm (1); For u = infinity, v = f = -50 cm which is the far point (1); -1 D: f = -100 cm (1); 1/u = 1/v - 1/f = -1/50 + 1/100 (1); u = -100 cm (1). | 1+1+1+1+1 |
| | **OR:** Presbyopia: weak ciliary muscles (1); Upper lens: concave for distance (1); Upper f = -2 m (-0.5 D) (1); Lower lens: convex for reading (1); Lower f = 33.3 cm (+3 D) (1). | 1+1+1+1+1 |
| 20 | Refraction as light enters the drop (1); Dispersion since n varies with colour (1); Internal reflection at the back (1); Sun must be behind the observer, low in the sky (1); Red bends least and lies at the top (1). | 1+1+1+1+1 |
| | **OR:** In dim light the iris dilates the pupil (1); In bright light it contracts the pupil (1); For near objects the ciliary muscles contract (1); Lens thickens, focal length decreases (1); For distant objects the opposite (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** slightly less | No advance sunrise or delayed sunset. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | No scattering in space. |
| 3 | **(b)** it is much closer and extended | Averaging effect. |
| 4 | **(d)** the far point of the eye is 50 cm | f = -50 cm. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | Presbyopia with myopia. |
| 6 | **(d)** bifocal lenses | Two corrections are needed. |
| 7 | **(c)** about 67 cm | 1/v = 1/40 - 1/25 = -3/200. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | The observer must stand between the Sun and the rain to receive the returned light. |
| 9 | **(d)** rod cells | Rods are light-sensitive. |
| 10 | **(d)** 2 m | f = -2 m. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | f = -1/3 m = -33.3 cm (1); Far point = 33.3 cm (1). | 1+1 |
| 12 | n is larger for violet (1); Violet travels slower in glass (1). | 1+1 |
| 13 | Light is monochromatic (1); Only one wavelength exists (1). | 1+1 |
| 14 | Slab faces are parallel (1); Prism faces are inclined, so the dispersion adds up (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Light refracts at the first surface and disperses (1); It reflects once inside at the back surface (1); It refracts again as it exits, with violet deviated the most (1). | 1+1+1 |
| 16 | Clear air has small molecules scattering mainly blue (1); Dust scatters all colours (1); So the dusty sky looks whitish (1). | 1+1+1 |
| 17 | Concave lens makes objects look smaller (1); Convex lens makes objects look bigger (1); Concave helps far vision (1). | 1+1+1 |
| | **OR:** v = 2 cm, u = -25 cm (1); 1/f = 1/2 + 1/25 = 27/50, f = 1.85 cm (1); Possible only if the lens can shorten its focal length to 1.85 cm (1). | 1+1+1 |
| 18 | The lens itself becomes cloudy (1); Spectacles cannot clear it (1); Surgery replaces the lens (1). | 1+1+1 |
| | **OR:** Rods work in dim light (1); Cones work in bright light and detect colour (1); Both convert light into electrical signals (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Cornea and lens correspond to the lens system (1); Iris and pupil correspond to aperture and diaphragm (1); Retina corresponds to the film or sensor (1); Ciliary muscles change the focal length, whereas a camera changes the lens position (1); Optic nerve corresponds to the cable carrying data (1). | 1+1+1+1+1 |
| | **OR:** Take two glass tanks, one with pure water and one with water and a few drops of milk (1); Pass a torch beam through each (1); Milk particles are colloidal and scatter the light (1); The beam is visible in the milk water (1); Pure water has no particles to scatter light (1). | 1+1+1+1+1 |
| 20 | Longer path scatters blue away so red remains (1); Rays from the lower and upper edges are refracted differently (1); The disc looks flattened (1); Atmospheric refraction bends rays over the horizon (1); Sun is seen about 2 minutes after the actual sunset (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Myopia: f = -125 cm (1); P = -0.8 D, concave (1); Hypermetropia: 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D, convex (1); Concave diverges for myopia; convex converges for hypermetropia (1). | 1+1+1+1+1 |

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